

BT 503 Final Current Solved – Mostly Question Solved from Internet

MCQS

Question and Answers

Feature of phenol that make it pollutant?

Phenols and some detergents may also mimic the activity of oestrogen.

Name enzyme equipment and solution used in genetic engineering?

the enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell. It is also essential to have some means of determining whether or not the transfer has been successful. This is achieved by the use of marker genes.

Importance of anaerobic zone?

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

Biological process of phosphorus removal activated sludge treatment?

The **process** of **removing phosphorus** by accumulating it with biomass is referred to as enhanced **biological phosphorus removal (EBPR) process**. The **phosphorus removal** efficiency for **biological** systems depends on the **phosphorus** content of the **sludge** removed and the efficiency of the solids separation **process**. **(Internet)**

When the biological excess sludge is wasted from the system, phosphorus is removed... For higher efficiencies of phosphorus removal, effluent polishing methods can adopted: • addition of coagulants (metallic ions): phosphorus precipitation • effluent filtration: removal of the phosphorus present in the suspended solids in the effluent • combination of the addition of coagulants and filtration. **(PPTS)**

Photodegradation and name its forms?

Photodegradation involves the **process** of oxidation (loss of electrons, increase in oxidation state) and hydrolysis (split of chemical bonds by adding water). ... **Photodegradation** does not talk about the ageing or degradation of materials with the use of heat or infrared light.

What are living conditions extremophile give example?

Three **examples** of **extremophiles** are Picrophilus torridus (a thermoacidophile adapted to hot, acidic **conditions**), Antarctic krill (a psychrophile), **and** the Pompeii worm (a thermophile).

What are application of biosensor in detection of biocides?

Biosensors have been extensively applied in clinical, food and environmental areas due to the advantages of fast **detection** speed, high selectivity and sensitivity (Malhotra and Chaubey, 2003). In this study, the contribution of bioluminescence and luminometer makes a **biosensor** for **detection** of water pollution.

Explain genomic libraries?

A **genomic library** is a collection of the total **genomic** DNA from a single organism. The DNA is stored in a population of identical vectors, each containing a different insert of DNA. ... The fragments are then inserted into the vector using DNA ligase.

Biofilter?

Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants. Common uses include processing waste water, capturing harmful chemicals or silt from surface runoff, and microbiotic oxidation of contaminants in air.

Biodiesel?

Biodiesel is a form of diesel fuel derived from plants or animals and consisting of long-chain fatty acid esters. It is typically made by chemically reacting lipids such as animal fat, soybean oil, or some other vegetable oil with an alcohol, producing a methyl, ethyl or propyl ester.

Plant associated with fungi?

Mycorrhizae: Associations of fungi with roots of vascular plants

mycorrhizae, mycorrhiza or mycorrhizas) is a symbiotic association between a **fungus** and a **plant**. The term mycorrhiza refers to the role of the **fungus** in the **plant's** rhizosphere, its root system. Mycorrhizae play important roles in **plant** nutrition, soil biology and soil chemistry.

Water system?

water system. a river with all its tributaries. a **system**, including reservoirs and mains, for storing and supplying **water** for use by a community.

How plant can use be reduced pollution?

Plants improve air quality through several mechanisms: they absorb carbon dioxide and release oxygen through photosynthesis, they increase humidity by transpiring water vapor through microscopic leaf pores, and they **can** passively absorb **pollutants** on the external surfaces of leaves and on the **plant** root-soil system.

How animal work as bioindicator two example?

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

Example of a **bioindicator** is lichens. These plants, which live on surfaces such as trees or rocks or soil, are very sensitive to toxins in the air.

Greasewood indicates saline soil

Define phytoremediation?

Phytoremediation /ˌfaɪtəʊrɪˌmiːdɪˈeɪʃən/ refers to the technologies that use living plants to clean up soil, air, and water contaminated with hazardous contaminants.

What are biodegradation?

Biodegradation is the breakdown of organic matter by microorganisms, such as bacteria and fungi.

Bacteriophage?

In bacteria, the most likely candidates for genetic transfer are plasmids and bacteriophages.

Biosensor advantage?

Advantages of tissues as **biosensors** include the following: easier to immobilize compared to cells and organelles. the higher activity and stability from maintain enzymes in natural environment.

Sources dioxin?

- **Combustion** sources, e.g. municipal waste or medical waste incinerators and private backyard barrel burning.
- Metal smelting.
- Refining and process sources.
- Chemical manufacturing sources.
- Natural sources

Method to get quality crop?

The average **quality** can be increased by planting seeds into fertilized soil. Increasing Farming Skill also increases the chance to harvest a **quality crop**. If a Seed Maker turns a higher-**quality crop** into seeds, the seeds are no more likely to produce higher-**quality crops** when grown and harvested.

PCB TNT and subtreated biofilter?

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Phase sludge water in sec wastewater treatment?

Sludge is a semi-solid slurry that can be produced from a range of industrial processes, from **water treatment**, **wastewater treatment** or on-site sanitation systems. ... Industrial **wastewater treatment** plants produce solids that are also referred to as **sludge**.

In which filed biotechnology used?

- manufacturing process, • waste management, • pollution control

Thermophile and its used in PCR? Thermophile and application taq polymerase in PCR?

In PCR: DNA Polymerase of *Thermus aquaticus* (Taq Polymerase), *Pyrococcus furiosus* (Pfu Polymerase). Clean energy: bioethanol production, hydrogen production fuel cells, etc.

Define Extremophiles and name three categories?

Extremophiles are animals that live and thrive under extreme environmental conditions

Psychrophiles can tolerate these circumstances.

Halophiles... survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds.

Acidophiles thrive in the conditions of low pH, typically below 5

Extremophile in nature?

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Trickling filter?

A **trickling filter** is a type of wastewater treatment system. It consists of a fixed bed of rocks, coke, gravel, slag, polyurethane foam, sphagnum peat moss, ceramic, or plastic media over which sewage or other wastewater flows downward and causes a layer of microbial slime (biofilm) to grow, covering the bed of media.

Advantage and disadvantage of incineration?

Advantages: • Drastic volume reduction, • Sterilisation

Disadvantages: • High costs, • Ash disposal, • Atmospheric pollution.

Effect of dioxin on human?

dioxins may result in skin lesions, such as chloracne and patchy darkening of the skin, and altered liver function. Long-term exposure is linked to impairment of the immune system, the developing nervous system, the endocrine system and reproductive functions

Long 5 Marks

Explain environmental factor bioremediation?

bioremediation is the use of living organisms, primarily microorganisms, to degrade the **environmental** contaminants into less toxic forms. It uses naturally occurring bacteria and fungi or plants to degrade or detoxify substances hazardous to human health and/or the **environment**.

Detail note on BCD and use of biosensor to detect these?

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Biodiesel advantage?

Biodiesel is a domestically produced, clean-burning, renewable substitute for petroleum diesel. Using biodiesel as a vehicle fuel increases **energy** security, improves air quality and the environment, and provides safety benefits.

Solid waste step?

1. Identify the types of waste.
2. Identify the sources of waste.
3. Determine the potential health hazards from waste.
4. Determine the volume of waste generated
5. Identify safe collection method/s.
6. Identify safe transportation method/s.
7. Identify safe disposal method/s.

Genetic modification in poplar plant used to remove mercury? Describe how?

Genetic modification of a poplar to enable mercury to be removed from the soil and converted to a form able to be released to the atmosphere.

mer A gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form.

Initially the gene was transferred to *Arabidopsis thaliana*. The gene was transferred by gene guns to poplar tree (*Liriodendron tulipifera*) embryogenic material. The resulting poplar plantlets were found to exhibit tolerance to mercury and to volatilise it at 10 times the rate observed in untransformed yellow poplar plantlets.

Factor affect bioremediation?

The factors that directly impact on bioremediation are energy sources (electron donors), electron acceptors, nutrients, pH, **temperature**, and inhibitory substrates or metabolites. One of the primary distinctions between surface soils, vadose zone soils and groundwater sediments is the content of organic material.

Properties of soil which has effect on pollution?

The application of waste or **polluted** water into the **soil** alters its physical and chemical, thereby affecting the growth of agricultural crops and other living organisms. Even the extensive agricultural practices such as fertilizer and pesticide application deteriorate the **soil quality**.

Detail note on environmental biotechnology?

Environmental biotechnology. **Environmental biotechnology** is the branch of **biotechnology** that addresses **environmental** problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

Explain recombinant microorganism especially yeast and microorganisms?

GE of microorganisms for use in environmental biotechnology: expansion of metabolic pathways either to modify the existent metabolic capability, or to introduce new pathways.

Many examples... A strain of *Escherichia coli* into which was engineered some 15 genes originating from *Pseudomonas*. These were introduced to construct a pathway able to produce indigo for the dyeing of denim (Bialy, 1997). The traditional method requires the use of toxic chemicals.

Long 10 Marks

Briefly describe category of solid waste?

Different categories of solid waste include:

- Non-combustibles: Metal, tin cans, bottles, stones, etc.
- Ashes/dust: Residue from fires used for cooking.
- Bulky waste: Tree branches, tyres, etc.
- Dead animals: Carcasses of domestic animals and livestock
- Hazardous waste: Oil, battery acid, medical waste.
- Construction waste: Roofing, rubble, broken concrete, etc.

Genomic Library?

Genomic libraries Collection of total Genomic DNA from an organisms. The genomic DNA is isolated, purified and cut up into pieces of a size suitable to be inserted into a cloning vector, following by ligation and transformation into a host cell. These DNA pieces may either be ligated as a total mixture, into a suitable vector to produce a genomic library, or a specific piece may be isolated and prepared as described ago. Genomic libraries are very useful, as they may be amplified, and accessed almost limitlessly, to look for a specific DNA Sequence. If the genomic library is of a eukaryotic origin the genes will contain non-coding regions, called introns. (Prokaryotes do not contain introns). This is a problem if the gene is to be expressed. This problem can be avoided by using cDNA.

Sludge treatment procedure?

Sewage **sludge treatment** describes the **processes** used to manage and dispose of sewage **sludge** produced during sewage **treatment**. **Sludge** is mostly water with lesser amounts of solid material removed from liquid sewage. ... Reducing **sludge** volume may increase the concentration of some of these toxic chemicals in the **sludge**.

Phytovolatilization?

Phytovolatilization is a process, in which plants take up contaminants from soil and release them as volatile form into the atmosphere through transpiration. The process occurs as growing plants absorb water and organic contaminants.

What is cloning factor? Why important in genetic engineering?

Cloning allows for the creation of multiple copies of **genes**, expression of **genes**, and study of specific **genes**. ... Plasmids have been highly **engineered** as vectors for molecular **cloning** and for the subsequent large-scale production of **important** molecules, such as insulin.

Southern blotting and western blotting?

In Southern blotting, DNA fragments are resolved by electrophoresis on a gel. These resolved fragments are then shifted to a membrane (nylon or nitrocellulose).

Western blotting: Proteins (e.g., antibodies antigens interaction)